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A Primer for Uncertainty Quantification for Neural Network Potentials

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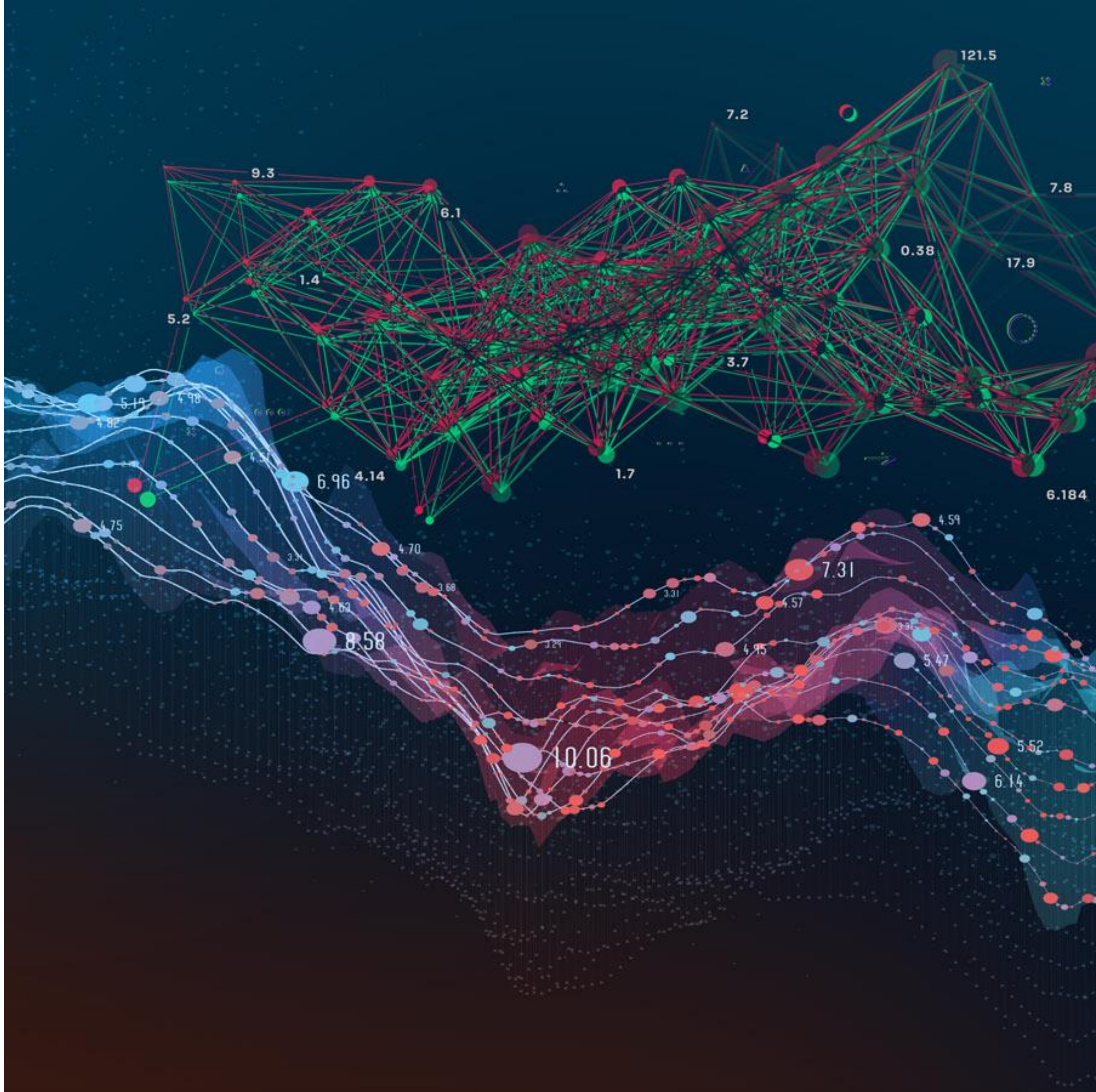

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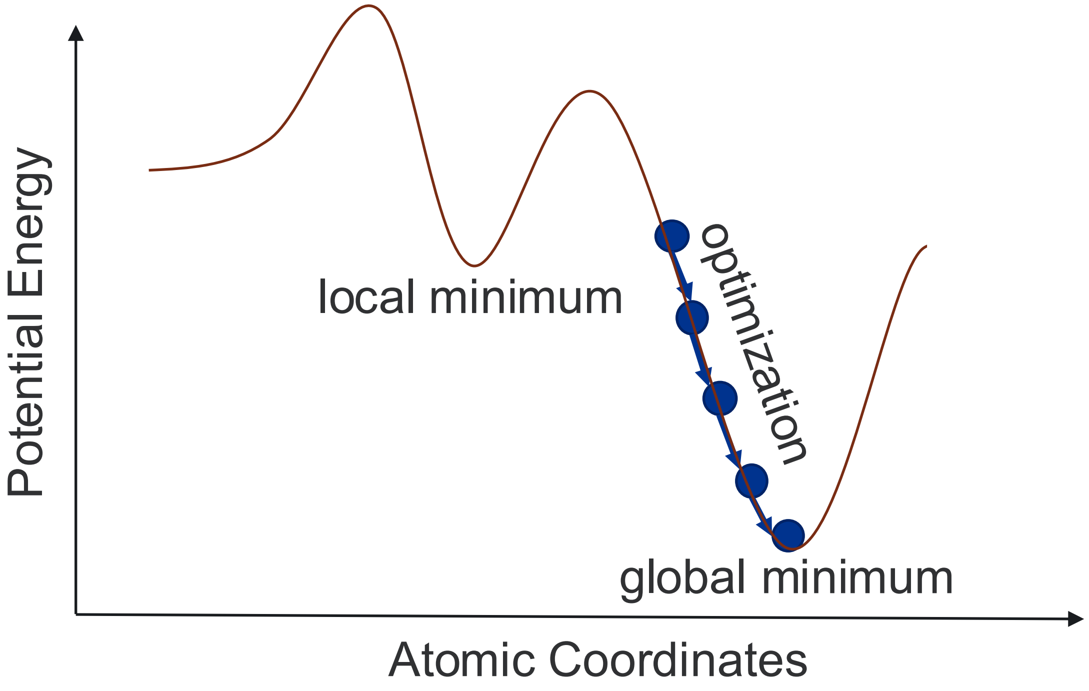
The Basics of Molecular Simulations

- Atomic simulations explore the potential energy surface (PES) of a molecule
- The PES is a function that relates **atomic positions** to a molecule's **potential energy**
- The negative gradient of the PES corresponds to the **atomic forces** on each atom
- Minima on the PES represent stable (physically observable) configurations



- During ***optimization***: simulations are performed to minimize the gradient over the PES to obtain stable minima
  - Many repeated optimizations are needed to locate the global minimum configuration
- During ***molecular dynamics***: simulations are performed over the PES at a set temperature to examine the "evolution" of the system
  - May sample multiple minima along with non-minima

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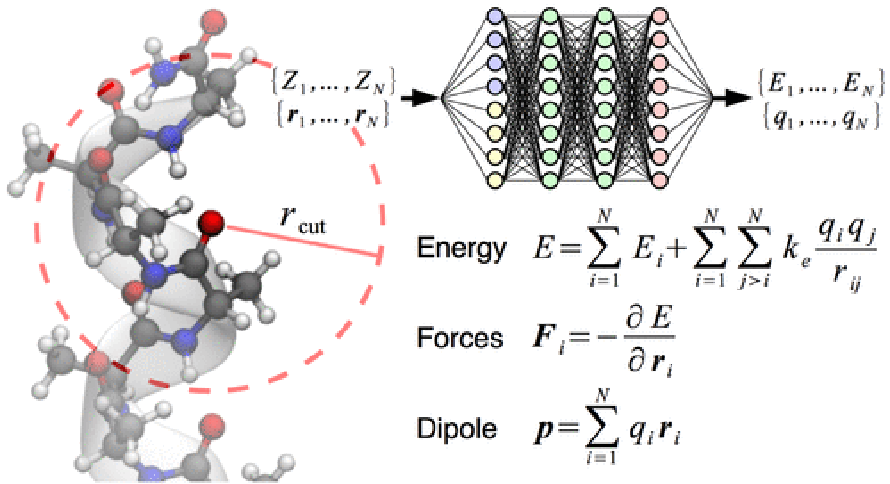
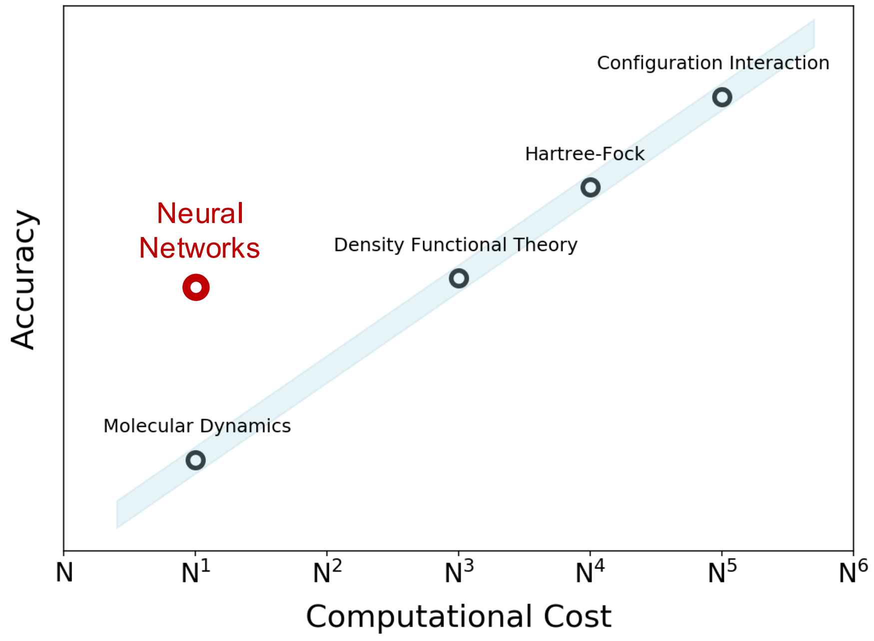
Simulations are Computationally Expensive!

Need:

- Computational speed-ups for dynamics simulations or geometry optimizations on large systems or over long times, while maintaining accuracy



Unke et al. *J. Chem. Theory Comput*. **15**, 6, 2019.

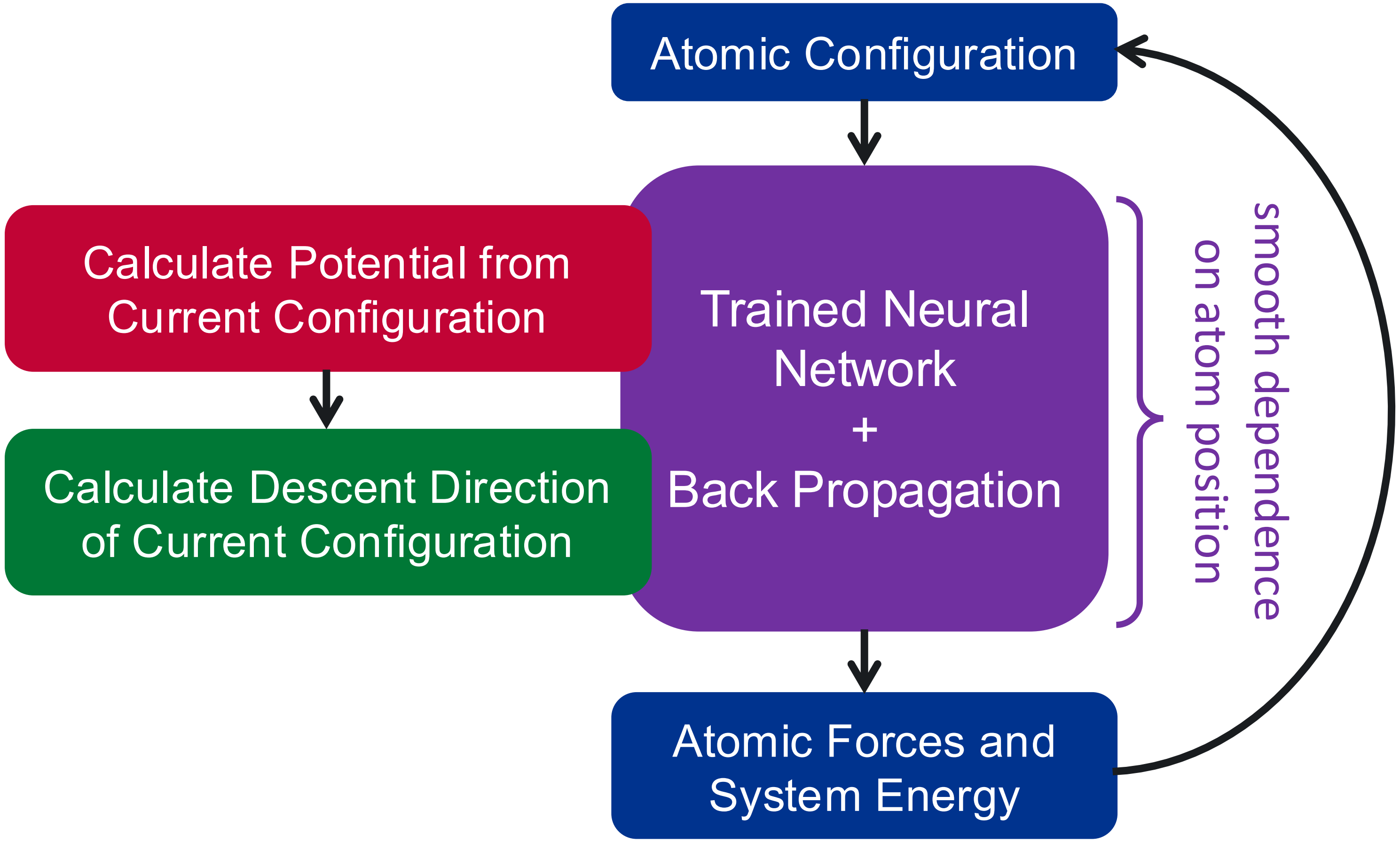


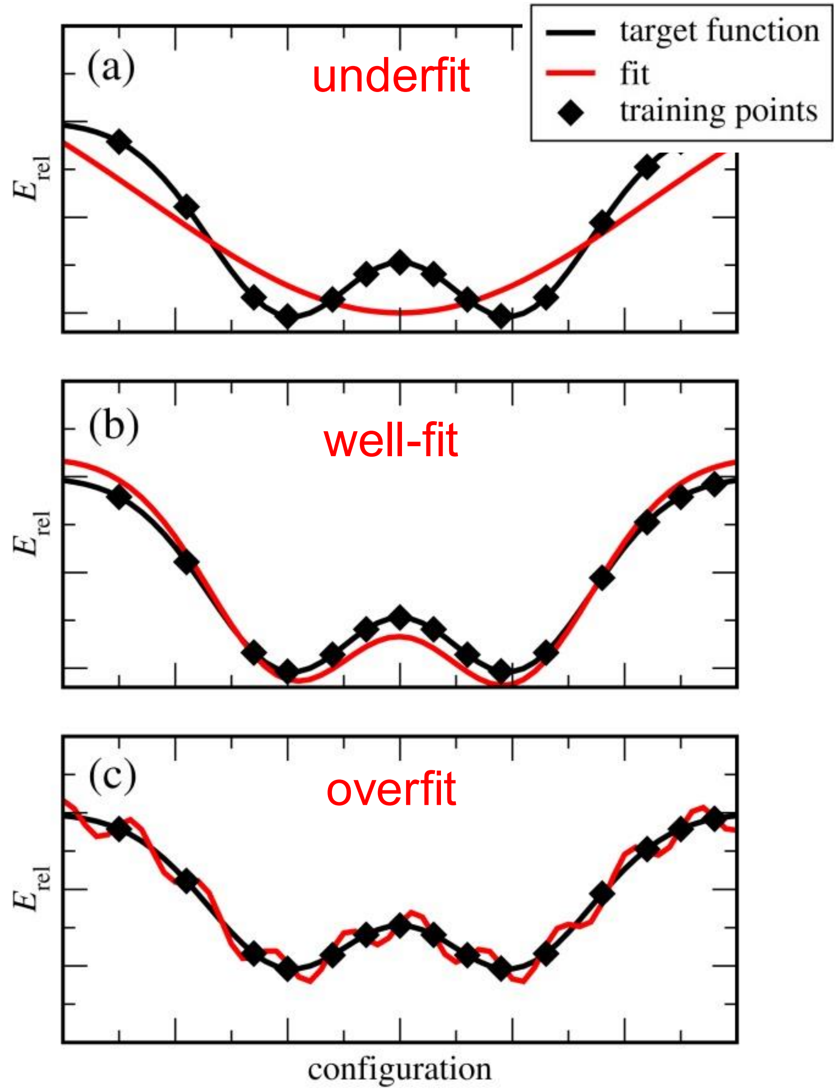
Solution:

- Develop a surrogate model that quickly estimates a molecule's energy and forces on each atom

Neural Network Potentials (NNPs)

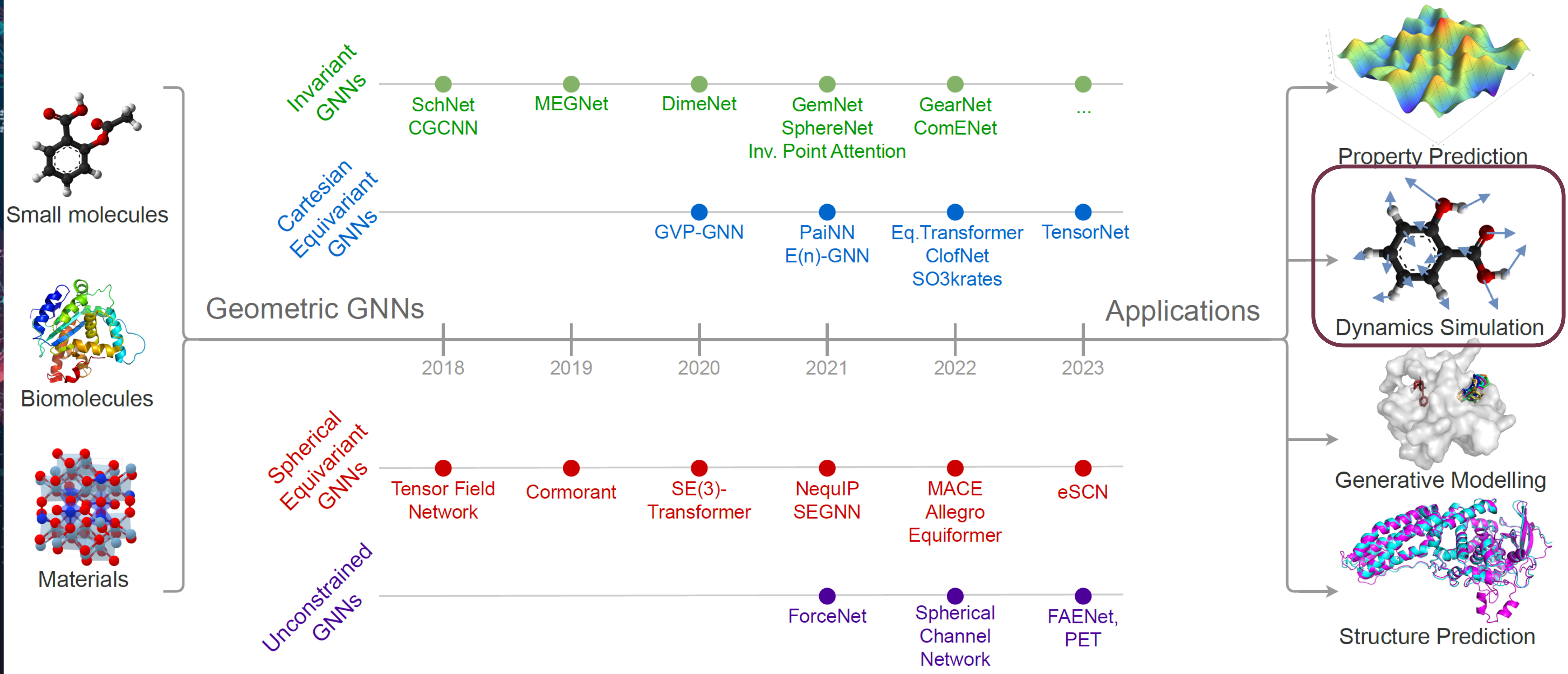
NNPs attempt to approximate the PES function





Behler. *Int. J. Quant. Chem.* **115**, 16, 2015.

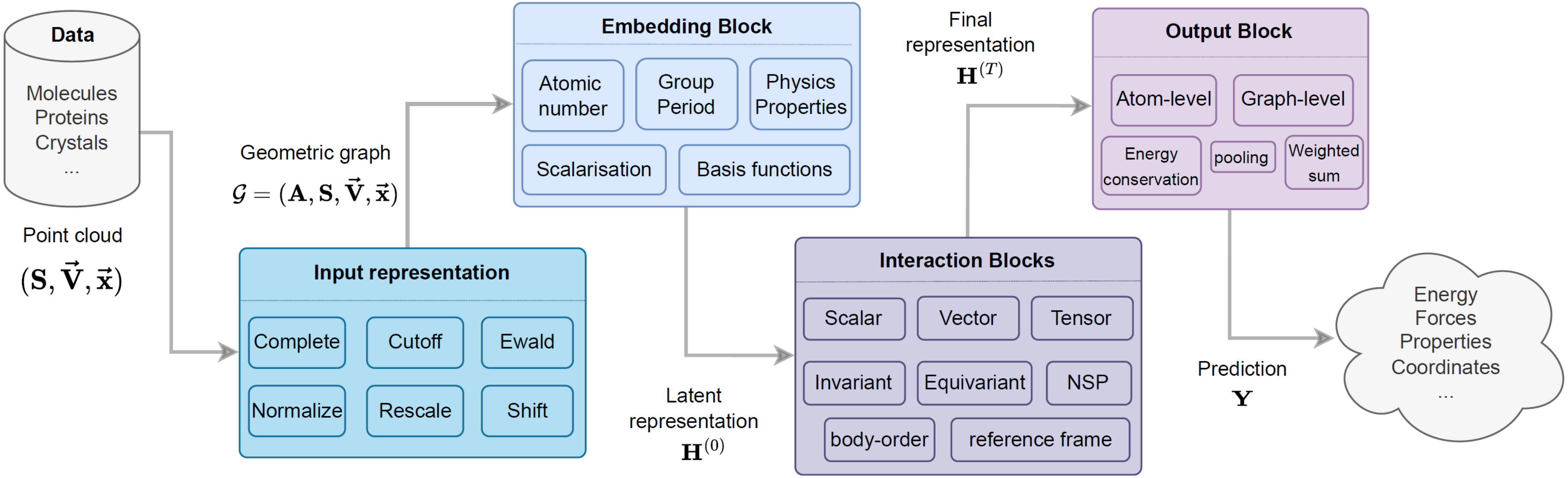
Class of Surrogate Models: Atomic Graph Neural Networks (GNNs)



"A Hitchhiker's Guide to Geometric GNNs for 3D Atomic Systems." (2023) arXiv:2312.07511

Common NNP Workflow

1. Latent representations are learned for each atom
2. Atom representations are updated by interactions between atoms
3. Contributions from each atom are pooled to give graph-level output



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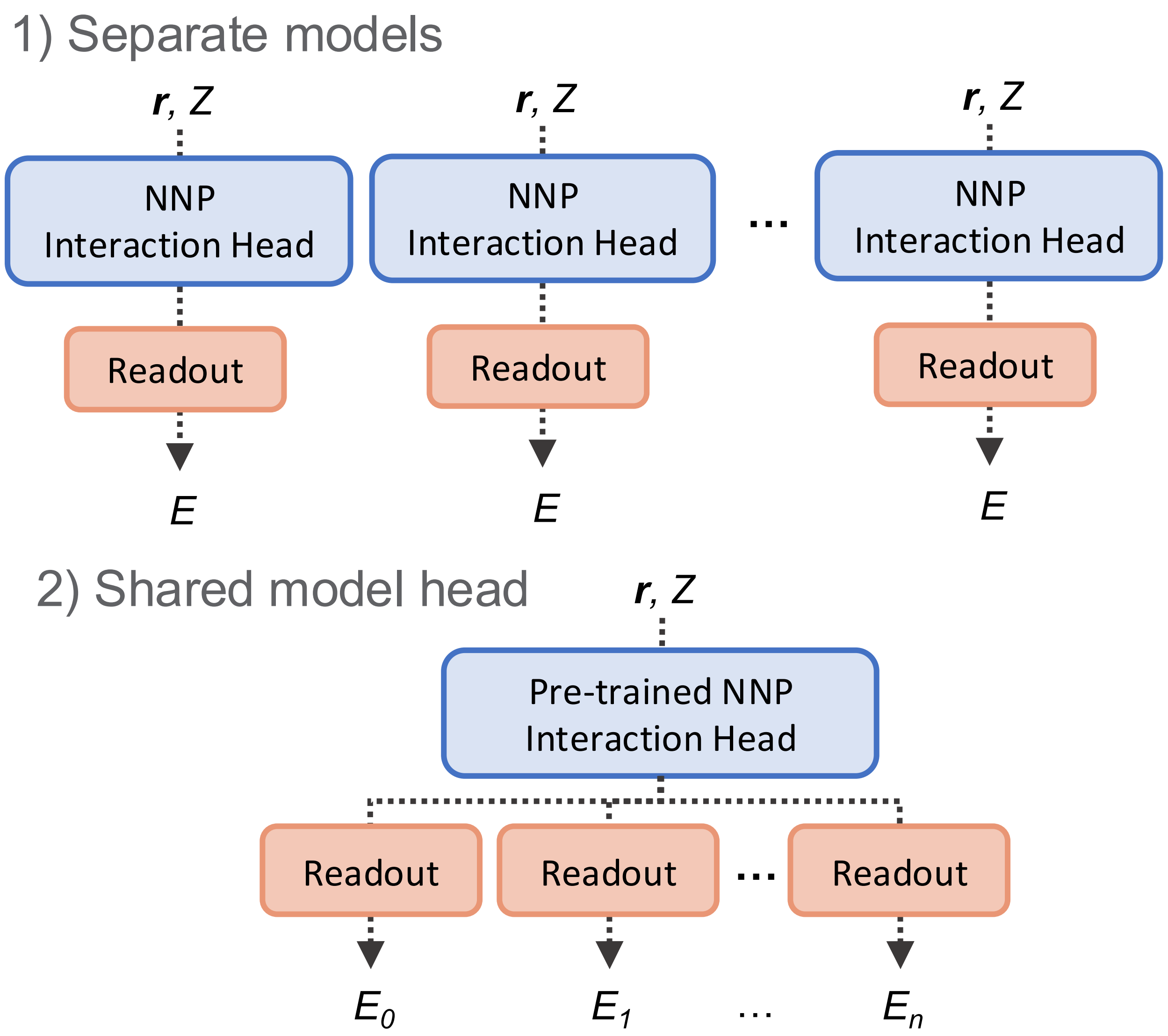
Model Ensembling

- A set of models are trained with different initializations, train:val splits, hyperparameters, etc.
  - Typically 7-10 models trained
- Std. dev. of models is used as uncertainty
- Often used for active learning to generate most useful training set



Quantile Regression

- Readout layers trained to produce upper and lower quantile of energy

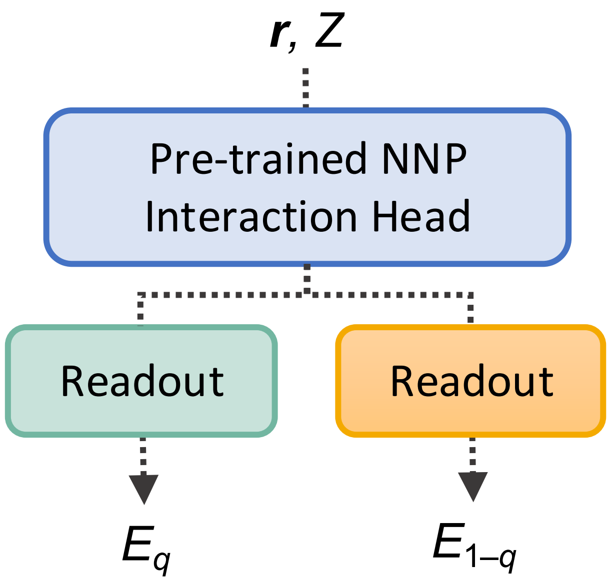
$$U = \frac{E_q - E_{1-q}}{2}$$

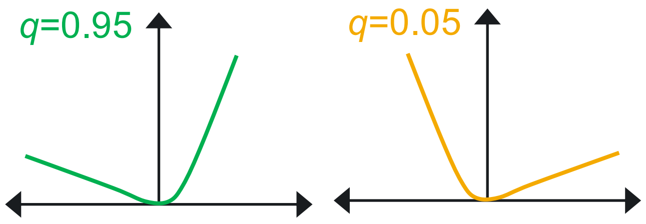
- Smooth pinball loss function
  - No extra info needed from simulations
  - Train against DFT energy (E)

$$L_{\text{total}} = L_q + L_{q-1}$$

$$L_q = q\left(E - E_q\right) + \frac{1}{\alpha}\log(1+\exp(\alpha(E_q - E)))$$

Single model head





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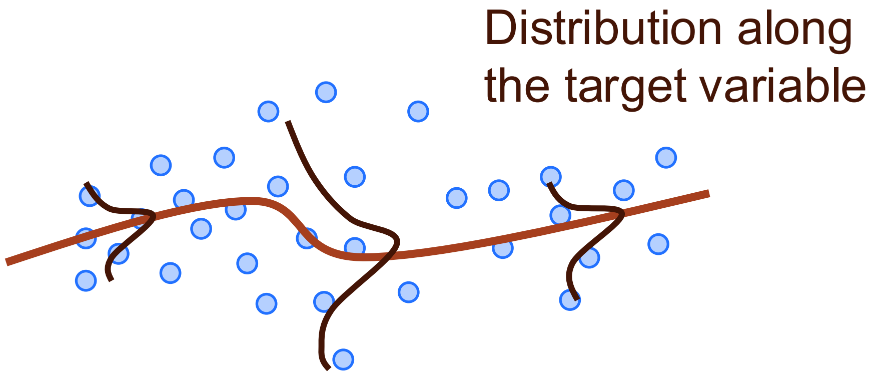
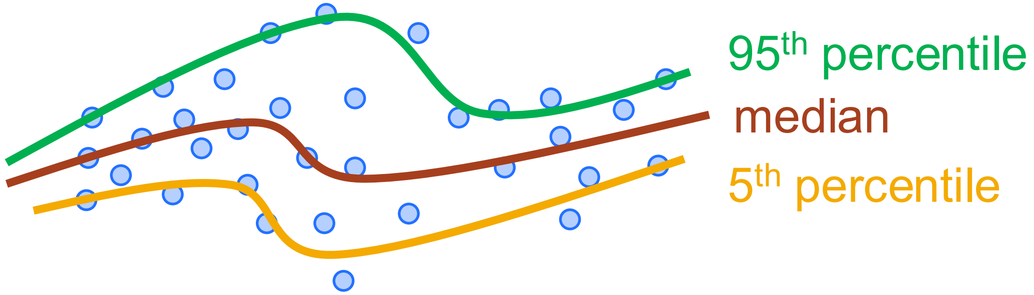
Quantile Regression

- Normally trained models give a point estimate, typically targeted towards the mean
- Point estimates don't reflect the distribution of values along the target value
- Quantile regression fits to percentiles
- Produces confidence interval (CI) instead of point estimate

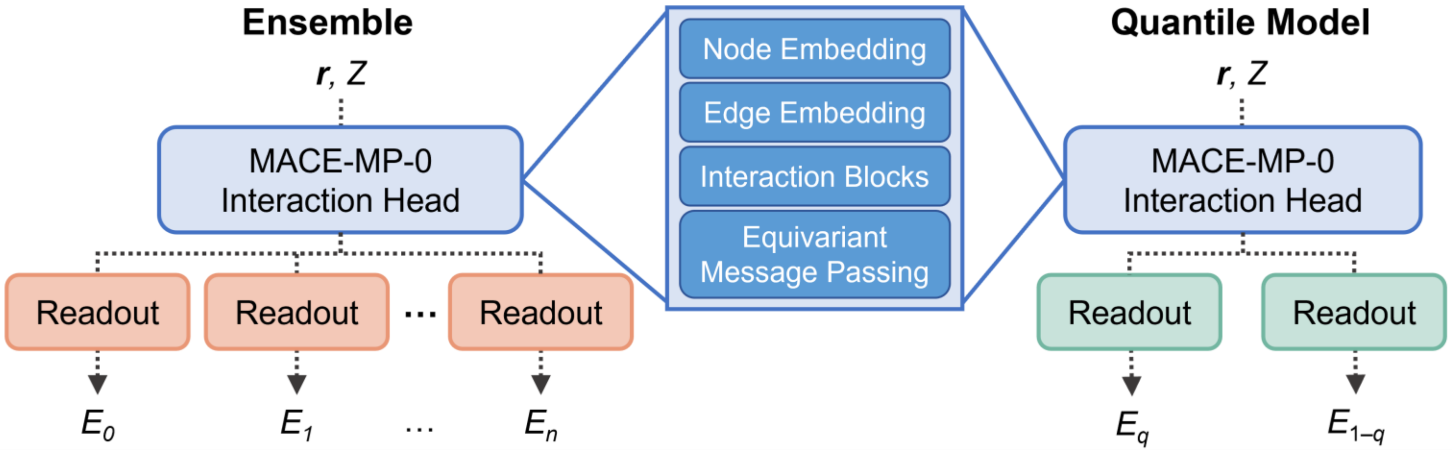




UQ for MACE-MP-0

Student's t-value for 90% CI

$$U = \frac{t * \sigma}{\sqrt{n}}$$



90% confidence interval (CI)

$$U = \frac{(E_{0.95} - E_{0.05})}{2}$$

- Training and Evaluation:
  - 7 models in readout ensemble
  - 1 quantile model
  - Unique 90k subset from MPtrj
  - Common 10k test set
  - *U* is half the length of the 90% CI

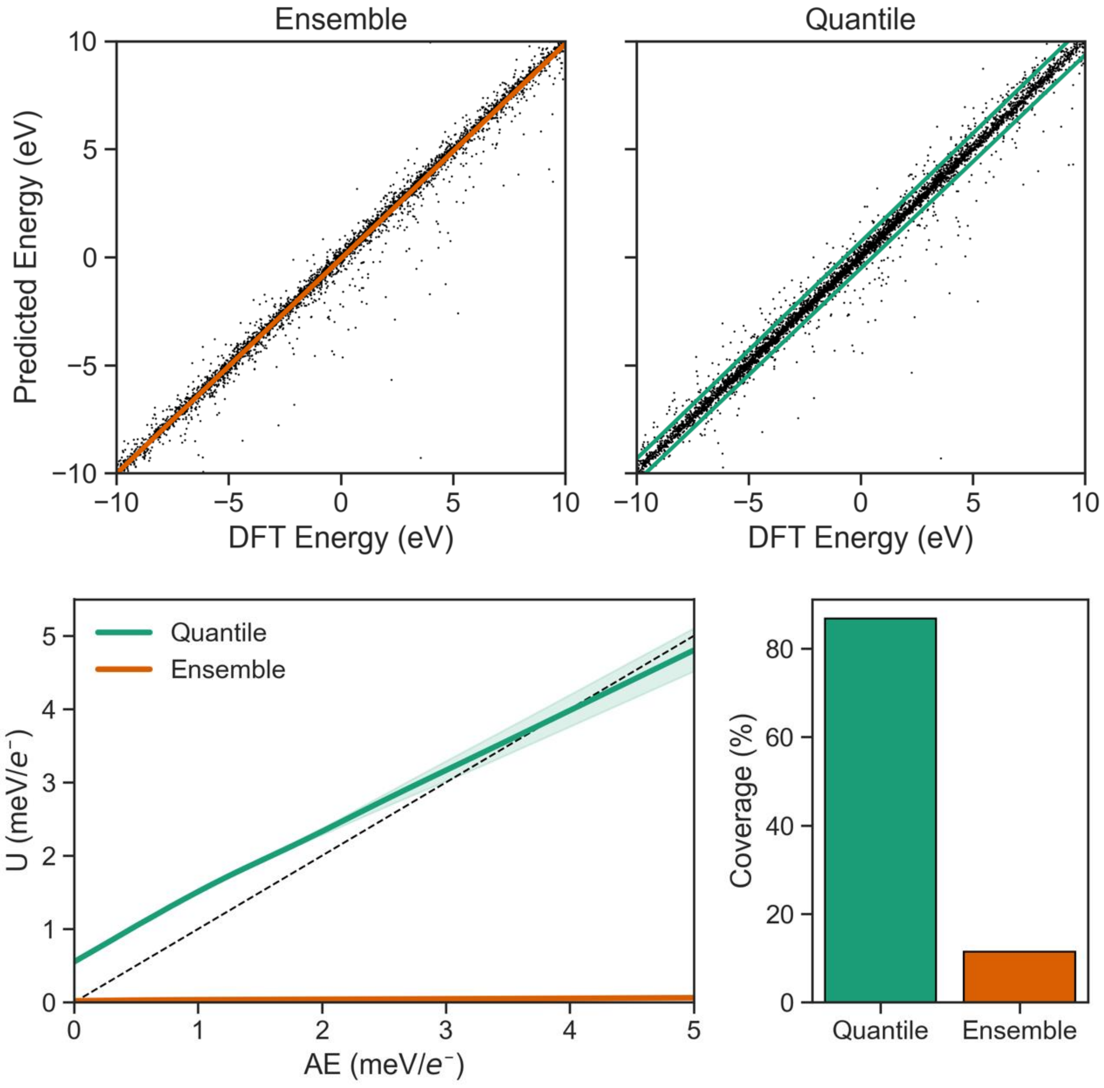
Test Set Performance

	Ensemble	Quantile
MAE *E* (meV/e⁻)	0.721	0.890
U (meV/e⁻)	0.036	1.391
Coverage	11%	87%

Bilbrey et al. (2025) *npj Computational Materials*.

Uncertainty's Relation to Prediction Error

- Ensemble is overly confident
  - Error bands include small fraction of samples
  - *U* only slightly increases with prediction error
- Quantile uncertainty is more representitative of prediction error
  - *U* increases at roughly the same rate as prediction error
  - Higher error samples show larger variance in *U*
  - For most – *not all!* – samples, *U* is on the same scale as error



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Uncertainty Quantification Recap

- During training, ensembling useful for active learning
  - Identifies out-of-distribution structures to add to training set
- Post-training, readout ensembling can be used to understand model (epistemic) uncertainty
  - Tells about goodness of model training
- Post-training, quantile regression can be used to understand data (aleatoric) uncertainty
  - Provides rough error calibration and indicates the overall expected error

	Readout Ensembling	Quantile Regression
Type of Uncertainty	Epistemic uncertainty from model variability	Aleatoric uncertainty from data
Source of Uncertainty	Model architecture and learned weights	Simulation choices – convergence criteria, XC functional, sampling method
Distribution Assumption	Predictions follow Gaussian-like distribution	None – distributions may be asymmetric
Computational Cost	Moderate – multiple models trained on subsets of full training data	Low – single model
Generalizability	Dependent on ensemble size	Dependent on data distribution

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Thank You!

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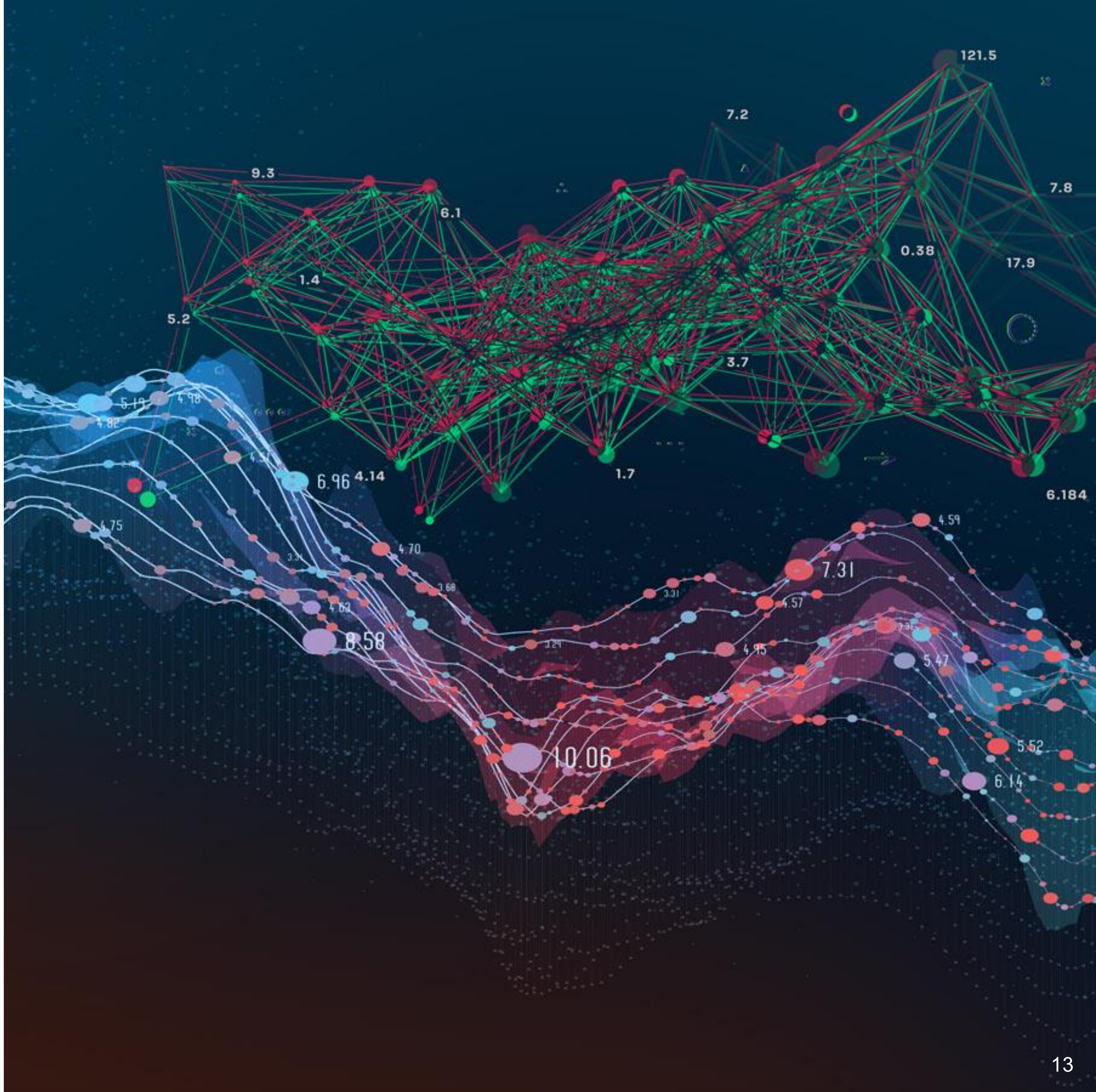

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